DSA4213 Assignment 1: Does better next-token prediction give better representations?

Language models, embeddings and downstream topic classification on WikiText-2 and AG News

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1 Question and setup

This report tests two common assumptions about small language models (LMs) on one corpus: that lower perplexity means a better model, and that better LMs give better text representations. The question is: *does better next-token prediction translate into better frozen representations for topic classification, and how much do input-embedding choices matter?* Part I trains four LM families on WikiText-2 under one recipe; Part II keeps the LSTM (the best full-history model of Part I and among the cheapest per epoch) and varies only its input table in a 3×2 (source × trainability) design; Part III feeds the frozen models' states to one linear classifier on AG News.

2 Data and common protocol

Corpus and vocabulary. WikiText-2 [1] (pre-tokenised, with its own `<unk>`) is lowercased once, one `<eos>` is appended per line, and the vocabulary is built from the train split only: 28,913 types. Lowercasing keeps the comparison with the uncased GloVe-6B fair; it shrinks the vocabulary from 33,278 types, so perplexities are not comparable with published numbers. Validation and test contain no out-of-vocabulary tokens beyond the dataset's own `<unk>` (6.2% of test tokens), which every model, including the n-gram, treats as an ordinary vocabulary item.

Streams and perplexity. Each split is one continuous token stream (train 2,075,678 tokens, validation 216,348); no padding exists, so nothing has to be excluded from any loss, and every model is scored on the same 244,102 test targets, perplexity being exp of the mean per-target cross-entropy. Recurrent models are scored with batch 1 and the state carried over the entire stream (full history); the Transformer with a sliding 128-token window moved by 32, scoring the last 32 targets of each window (the first window scores all 128; every later target sees $\geq$96 tokens of context). A second protocol, non-overlapping 128-token blocks with the state reset, is identical for every model; it matches the Transformer's training condition but is a cold start for the recurrent models (trained with the state carried), so their drop under it is an upper bound on the value of context beyond 128 tokens. Validation during training uses the same protocols.

Models and recipe. All neural models share a 300-d input embedding (the largest width GloVe-6B ships, so Part II can reuse it) initialised $\mathcal{U}(-0.1, 0.1)$ and a $300 \times V$ output layer, so they differ only in the sequence core: a 2-layer tanh RNN (0.36 M core parameters), a 2-layer LSTM [2] (1.44 M), a 4-layer pre-LN Transformer [3] with 6 heads and FFN width 1200 (4.37 M) and, as a parameter-count control, the same Transformer with 1 layer (1.12 M). The n-gram baseline is interpolated Kneser–Ney [4, 5] with one discount per order, orders 2–5 fitted and the order chosen on validation. The neural recipe is identical for every run (hyper-parameters in Appendix C): batch 32 × 128-token BPTT chunks with the state detached between chunks, dropout 0.3 (the Transformer additionally drops attention and FFN activations at 0.3), Adam (10^{-3}, 200-step warm-up), learning rate halved after two consecutive epochs without validation improvement, early stopping after three, at most 12 epochs, global gradient clipping at 1.0 with the pre-clip norm logged at every step. Unit tests check causality (future tokens cannot change earlier logits) and that input and output tables are untied.

Cost measurement and generation. All runs use one Apple M4 (16 GB) through PyTorch 2.14's MPS backend [6], one process at a time; timers synchronise the device, and epoch 1 (shader compilation) is excluded from per-epoch averages. Model size is the parameter count and the fp32 checkpoint. The common inference workload is (A) scoring the test stream in non-overlapping 128-token blocks, 16 per batch, with each model's own protocol reported alongside, and (B) generating 50 tokens for each of 5 prompts at batch 1 with top-k sampling (Transformer without KV cache). MPS is not bit-deterministic: the 24-epoch runs repeat the first 12 epochs of the seed-0 runs and agree at epoch 12 within 0.04 (LSTM; up to 2.5 PPL at intermediate epochs) and 0.00 at every epoch (Transformer); Part II's seed-1 repeats measure seed variance (about one point); `-device cpu` reproduces exactly. Prompts are the first 10 tokens of five seeded-random *validation* lines ($\geq$30 tokens, not headings, no `<unk>`); every model continues each for 50 tokens with greedy decoding and with top-k=40, T=1

Model	Test PPL	Test, ctx$\leq$128	Val, ep. 3	Val, $\leq$8 min	Val, ep. 12 (best)	Test, 24 ep.
KN 5-gram	214.9	215.5	–	–	227.4	–
RNN	173.0	177.2	219.6	210.0	187.3	–
LSTM	134.2	150.4	237.3	214.8	145.4	124.6
Transformer (4L)	140.9	146.4	227.9	227.9	150.1	132.3
Transformer (1L, control)	171.4	174.7	256.9	234.0	183.1	–

Table 1: Part I prediction quality on the common 244,102 test targets. ctx$\leq$128: state reset every 128 tokens for every model (the matched comparison); epoch-3 / $\leq$8-min columns: matched updates / wall-clock; last column: a separate 24-epoch run.

Model	Params (M)	Core (M)	Size (MB)	Train (min)	Score, blocks (tok/s)	Score, own (tok/s)	Gen. (ms/tok
KN 5-gram	5.8 M entries	–	156	0.1	370,565	346,238	18.5
RNN	17.7	0.36	71	23.2	46,392	6,779	1.6
LSTM	18.8	1.44	75	22.1	63,206	26,333	1.7
Transformer (4L)	21.8	4.37	87	29.8	50,500	12,694	3.4
Transformer (1L, control)	18.5	1.12	74	22.0	74,059	18,758	2.3

Table 2: Part I cost on one Apple M4 (MPS). Core: excluding the shared embedding and output layers; train: minutes to the best checkpoint (n-gram: the order-5 fit); score: throughput on the common workload (128-token blocks, 16 per batch) and under each model's own protocol (batch 1 for recurrent models); generate: batch-1 sampling.

sampling from one shared CPU sampler, so decoding is matched exactly across the n-gram and neural models; continuations are summarised by distinct-n ratios and the fraction of 4-grams repeating an earlier 4-gram of the same continuation.

3 Part I: four language models on one corpus

Observations. *(a) Prediction quality* (Table 1). With full history the order is LSTM (134.2) $<$ Transformer (140.9) $<$ RNN (173.0) $\approx$ 1-layer Transformer (171.4) $<$ KN 5-gram (214.9). Under the matched protocol (context reset every 128 tokens for every model, the common evaluation setup) the order of the two best models *reverses*: the Transformer loses only 5.6 points, the LSTM loses 16.2 and ends up behind it. *Where the differences sit* (Fig. 2). On the 3.4% of test targets seen at most 10 times in training the n-gram is the best model (12.25 nats vs. 13.16 for the LSTM); every neural model wins on words seen $\geq$11 times, up to the most frequent bucket ($>$10k: LSTM 2.26 vs. n-gram 2.65). The Transformer is the best model on `<eos>` (1.09 vs. 1.30) but worse than the LSTM and even the n-gram on `<unk>` (3.51 vs. 3.19 and 3.35).

(b) Cost (Table 2). The 4-layer Transformer needs 29.8 min of training against 22.1 for the LSTM and reaches a worse full-history perplexity with 3× the core parameters. On the common inference workload (Table 2) the LSTM is the fastest neural scorer; under their own batch-1 protocols the recurrent models lose 2–7× in throughput and the sliding window halves the Transformer's. Sampling costs 1.7 (LSTM) vs. 3.4 ms per token (Transformer, no KV cache). The n-gram fits all four orders in 10 s and scores at 346,238 tokens/s on the CPU, but in this dictionary implementation its 156 MB of tables exceed any checkpoint and sampling costs 18.5 ms per token.

(c) Learning behaviour (Fig. 1). All four neural models were still improving at the 12-epoch cap: no learning-rate halving or early stop fired, and the train/validation gap grows (LSTM: 113.1 with dropout on vs. 145.4 at epoch 12) without the validation loss turning up, i.e. no overfitting at this budget. Per epoch the LSTM and 4-layer Transformer track each other; per wall-clock minute the LSTM leads the Transformer. After three epochs the tanh RNN is the *best* model (219.6 vs. 237.3 and 227.9); it learns fastest and saturates first.

(d) Generated text (all continuations in Appendix A, repetition statistics in Table 6). Under greedy decoding the RNN and 1-layer Transformer loop (74% and 57% of 4-grams repeat an earlier one; n-gram 58%), the LSTM and 4-layer Transformer far less (25%, 31%). Top-k sampling removes repetition for every model ($\leq$2.6%), so sampled continuations differ in local fluency and consistency instead: the Transformer holds a topic across the 50 tokens (a "route description" heading after the highway prompt, a Jin/Song court after the 1644 prompt), the LSTM holds a looser one (military and geographic vocabulary after prompts 1 and 4), the RNN drifts within a sentence and the n-gram within a few tokens.

Explanations, separated from the observations. *The LSTM's lead is consistent with a context-length effect.* Under context reset the Transformer is the better predictor, so the LSTM's full-history lead is attributable (up to the cold-start caveat of §2) to information carried beyond 128 tokens (WikiText articles are thousands of tokens long); whether a longer-window Transformer would close the gap was not tested, and carrying state indefinitely is itself an architectural property. *Parameter count does not explain the ranking.* The 1-layer Transformer has about the LSTM's core size (1.12M vs. 1.44M) and is 37.2 points worse, while the 4-layer model has 3× the LSTM's core and is still worse with full history, so parameter count does not explain the ranking. *Training budget conditions every conclusion.* Nothing converged in 12 epochs and the ranking at epoch 3 (RNN first) differs from that at epoch 12, so we re-ran the LSTM and Transformer for 24 epochs (Table 1, dashed in Fig. 1): the LSTM early-stopped (best at epoch 18, test PPL 124.6), the Transformer was still improving at epoch 24 (132.3), and under context reset the Transformer still leads (137.3 vs. 141.0), so both conclusions survive a

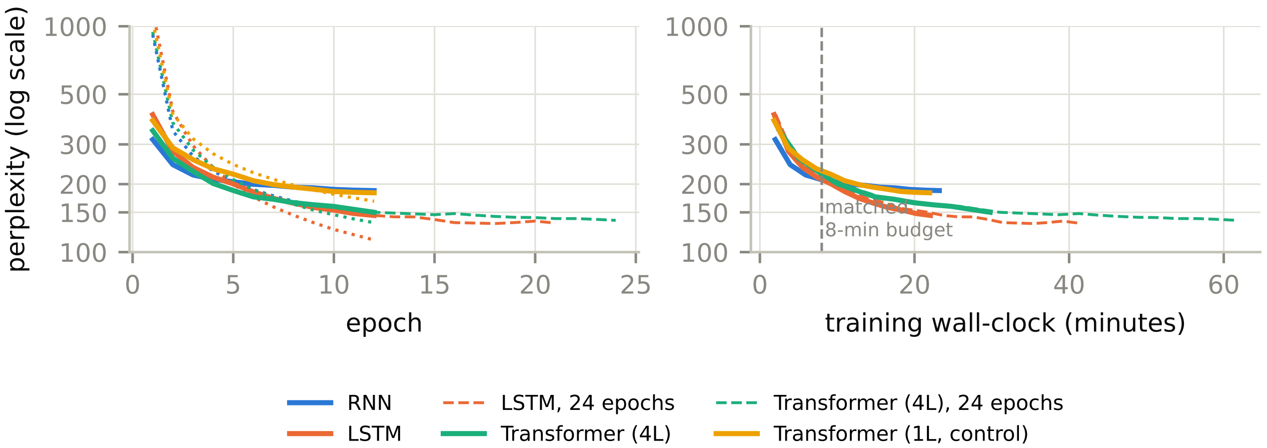


Figure 1: Validation perplexity (solid) and train perplexity (dotted) per epoch (left) and validation perplexity against training wall-clock (right); dashed: the separate 24-epoch runs.

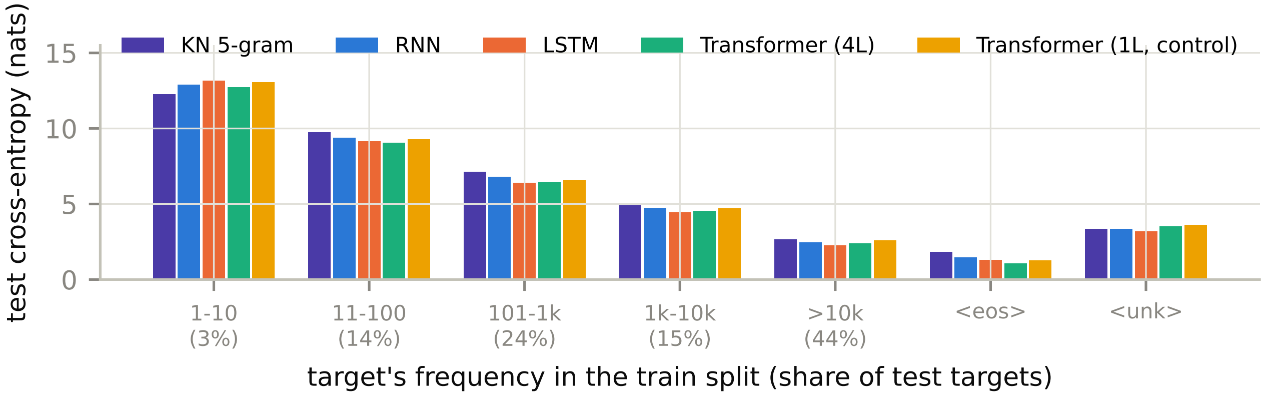


Figure 2: Test cross-entropy split by how often the target word occurs in the train split.

doubled budget. *The n-gram's edge on rare words* is expected from Kneser–Ney's continuation counts, which model how many contexts a rare type appears in, whereas a row whose word occurs at most ten times per epoch receives few informative gradient signals (a hypothesis; Part II tests its input side with pre-trained rows, while the equally under-trained output rows stay untested). *The RNN's slow batch-1 scoring* is most likely a kernel effect rather than an architectural one (batched in 128-token blocks it reaches 46,392 tokens/s, still below the LSTM), a hypothesis not tested on other hardware. *Greedy loops* track model quality among the four neural Part I models (74% and 57% vs. 25% and 31%) under one identical decoder, so there they reflect weaker modelling of the recent context rather than the decoding rule; the n-gram (58%) and the Part II cells (§4) show the relation is not monotone in perplexity.

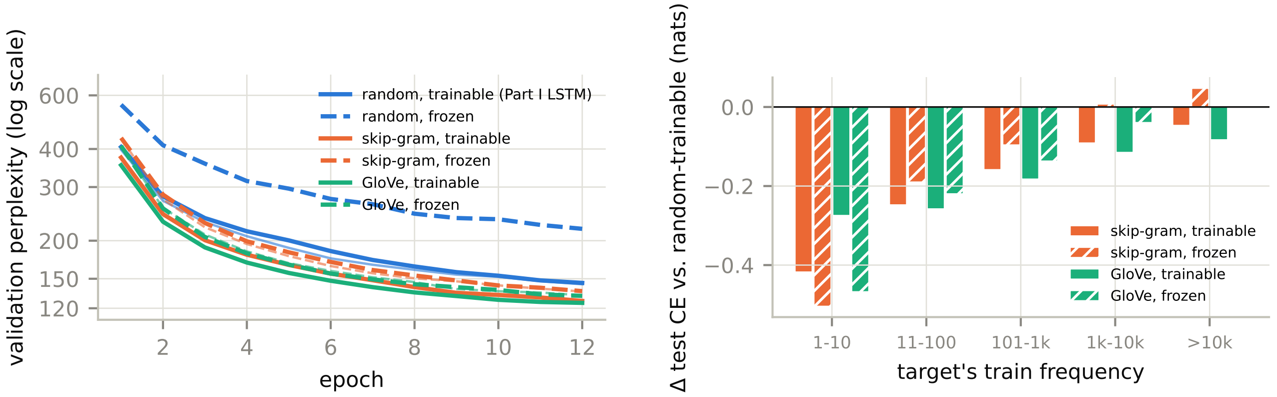
4 Part II: input-embedding ablation on the LSTM

Design. The LSTM of Part I is retrained with its 300-d input table replaced by (b) skip-gram vectors [7] trained by us on the WikiText-2 train stream only (gensim [8], negative sampling; 1.9 min on the CPU, reported separately from LM training) and (c) public GloVe-6B vectors [9], both kept entirely frozen; (a) is the Part I model itself. Because the required three cells confound *trainability* with *information content*, three extra cells complete a 3×2 factorial: the two pre-trained tables left trainable, and a frozen random table (no information at all). Skip-gram covers every type in the stream (`min_count`=1). GloVe covers 96.2% of the types and 94.7% of train tokens; the 1,104 missing types include the most frequent tokens of all (`<eos>`, `<unk>`, `@-@`) and are initialised $\mathcal{N}(0, \sigma_d)$ with σ_d the per-dimension standard deviation of the found rows. The tables differ in scale (mean row norm 6.47 GloVe, 2.37 skip-gram, 1.0 random), which a control run below tests. Freezing is verified by hashing the table before and after training; the required cells are run with seeds 0 and 1.

Observations. *Prediction quality* (Table 3). Both frozen pre-trained tables beat the jointly learned random table: skip-gram 127.9 and GloVe 123.8 against 134.0 test PPL, while the frozen random table collapses to 198.0. Letting the pre-trained tables train improves them further, to 119.0 and 116.5, the two best LMs here. Seed differences in the three repeated cells are 0.5, 1.1 and 0.8 points (noise floor about one point; the extra cells are single-seed); every contrast below is several times larger except skip-gram vs. GloVe among trainable tables (2.6 points).

Embedding source	Trainable	Cov. (%)	Emb. cost	Test PPL	Val ep. 3	Val ep. 6	$\|g\|$ mean	Clipped	Row norm
random	yes	100.0	–	134.0 ± 0.3	237.3	184.9	0.27	0.3%	1.46
random	no	100.0	–	198.0	358.1	274.2	0.32	0.4%	1.00
skip-gram	yes	100.0	1.9 min	119.0	200.8	155.9	0.27	0.3%	2.71
skip-gram	no	100.0	1.9 min	127.9 ± 0.5	228.6	170.4	0.29	0.3%	2.37
GloVe	yes	96.2	public	116.5	190.1	147.7	0.27	0.3%	6.57
GloVe	no	96.2	public	123.8 ± 0.4	205.7	156.0	0.28	0.3%	6.47
GloVe, rescaled to norm 1	no	96.2	public	135.4	258.8	184.6	0.33	0.4%	1.00

Table 3: Part II: the LSTM of Part I with six input-embedding settings plus a row-norm control (rows 1, 4, 6 required; test PPL ± half-range over seeds 0/1 where run, other columns seed 0; cov.: types with a pre-trained vector; $\|g\|$: mean pre-clip gradient norm; row norm: after training).



(a) Validation curves (thin: seed 1).

(b) Change in test CE vs. random-trainable by target frequency (hatched: frozen); the frozen random table, worse by 0.03–0.64 nats on every bucket, is omitted for scale.

Figure 3: Part II learning behaviour and where each embedding setting helps or hurts.

Convergence (Fig. 3a, columns "val ep. 3/6"). The ordering is visible after three epochs (Table 3), and the frozen skip-gram and GloVe cells pass the random cell's final perplexity at epochs 10 and 8; the random-frozen cell never does.

Stability. Mean pre-clip gradient norms (0.27–0.32) and clipped-step shares (0.3–0.4%) are indistinguishable across cells; the only differences are isolated spikes (maximum norm 13.1 in one seed of the random cell, 5.8 for the frozen random table, ≤2.3 elsewhere).

Generalisation. Frozen tables fit the training stream less but generalise relatively better: validation–train gaps at epoch 12 are 13.7 and 14.7 (frozen GloVe, skip-gram) against 32.2 for the random table and 26.4/29.8 once the pre-trained tables train, consistent with 8.7 M fewer trainable parameters.

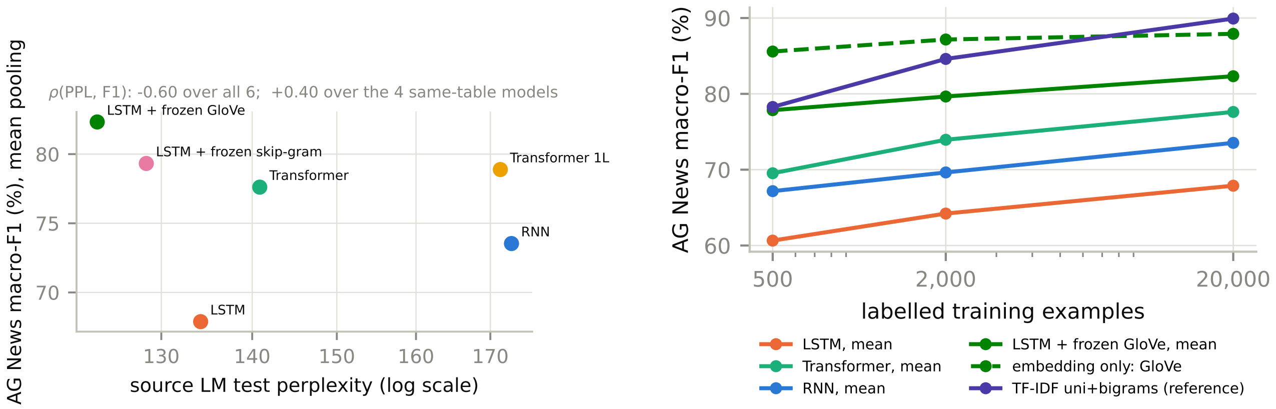
Where the gain sits (Fig. 3b). The per-token benefit of a frozen pre-trained table is largest on rare words: on targets seen at most 10 times, frozen GloVe and skip-gram change the cross-entropy by -0.47 and -0.50 nats relative to the random table; the change shrinks with frequency and vanishes on words seen more than 10k times (0.00 and +0.05). Rare targets are only 3.4% of the test set, so most of the total gain comes from words seen 11–1k times (74% vs. 19% of frozen GloVe's net gain). Trainable tables change less on the rarest words (GloVe -0.27) but are the only settings that also improve the most frequent ones (-0.08 and -0.05); the frozen random table loses on every bucket.

Generated text (Appendix A). Sampled continuations from the pre-trained cells are locally fluent WikiText, whereas the frozen random table degenerates into `<unk>`-laden text, at worst `<unk> @-@ <unk>` chains (top-k distinct-2 0.84 vs. 0.95). Greedy repetition is *not* monotone in perplexity: the best cells loop more (46% for trainable GloVe) than the Part I LSTM (25%), which these experiments do not explain.

Trainability versus extra data. The factorial separates the two factors the brief asks about. *Trainability* (frozen vs. trainable at a fixed source) matters enormously when the table carries no information (64.1 points) and only moderately once it does (skip-gram: 8.8; GloVe: 7.3). *Information in the table* (random vs. skip-gram at fixed trainability) is worth more: 14.9 points among trainable tables and 70.2 among frozen ones. Practically, a frozen skip-gram table trained on the same 2 M tokens in 1.9 min (under a tenth of one LM run) beats the trainable random table by 6.1 points. *Extra data* (GloVe's 6 B tokens against our 2 M) buys a further 4.1 points frozen and 2.6 trainable, less than having any sensible pre-trained table. Caveats pull GloVe both ways: its most frequent missing types (`<eos>`, `<unk>`, `@-@`) sit frozen at random vectors, a handicap, while GloVe-6B's Wikipedia 2014 plus Gigaword training data almost certainly contains the WikiText-2 source articles; and the row-norm control (last row of Table 3) shows that *input scale* is a large part of the frozen-table gains: the GloVe table rescaled to the random initialisation's norm (identical directions, 0.15× the length) reaches 135.4 test PPL, 11.7 points worse than the unscaled table and no better than the trainable random table. A frozen table fixes the input scale, which a trainable table must learn (the random table's mean row norm grows only from 1.0

Representation	LM test PPL (seed 0)	Acc.	F1, n=20,000 (±95% CI)	F1, n=2,000	F1, n=500
LSTM, last state	134.2	62.7	62.3 ± 1.0	59.4	54.5
LSTM, mean	134.2	68.1	67.9 ± 1.0	64.2	60.6
Transformer, last state	140.9	73.6	73.6 ± 1.0	69.9	66.6
Transformer, mean	140.9	77.7	77.6 ± 0.9	73.9	69.5
RNN, mean	173.0	73.6	73.5 ± 1.0	69.6	67.2
Transformer 1L, mean	171.4	79.0	78.9 ± 0.9	74.8	70.9
LSTM + frozen skip-gram, mean	128.4	79.4	79.3 ± 1.0	77.1	75.2
LSTM + frozen GloVe, mean	123.4	82.4	82.3 ± 0.8	79.6	77.8
table only: LSTM (learned)	–	75.2	75.1 ± 1.0	70.0	63.3
table only: RNN (learned)	–	79.4	79.4 ± 0.9	75.6	72.4
table only: Transformer (learned)	–	80.3	80.3 ± 0.9	77.0	72.9
table only: Transformer 1L (learned)	–	80.5	80.5 ± 0.8	76.8	73.6
table only: skip-gram	–	83.4	83.4 ± 0.8	81.1	80.1
table only: GloVe	–	87.9	87.9 ± 0.7	87.2	85.6
TF-IDF uni+bigrams	–	89.9	89.9 ± 0.7	84.6	78.3

Table 4: AG News test accuracy and macro-F1 (%) per frozen representation (300-d; TF-IDF 98,555-d) for 20,000 / 2,000 / 500 labels; C chosen on 2,000 validation documents; LM features from seed-0 checkpoints; CI: 95% bootstrap over the test set.



(a) Source-LM perplexity vs. downstream macro-F1.

(b) Macro-F1 against labelled-set size.

Figure 4: Does better next-token prediction give better representations? (a) Negative ρ means lower perplexity with higher F1.

to 1.46 in 12 epochs), and the skip-gram table (norm 2.4) presumably benefits in the same way; downstream, where features are standardised, scale cannot matter, so Part III's GloVe advantage is content. Our reading, a hypothesis: with 2 M tokens and 12 epochs the 8.7 M-parameter input table is the slowest-learning part of the model (rare rows receive few updates, and Adam's per-parameter steps grow the table's scale slowly), so giving every row a usable, well-scaled vector before LM training is worth more than the freedom to adjust it during training.

5 Part III: representations for topic classification

Setup. Each AG News [10] document (title + description, tokenised with sacremoses [11] to the WikiText conventions, lowercased, mapped to the WikiText vocabulary at 92.5% token coverage, `<eos>`-prefixed, truncated at 96 tokens for 0.7% of documents) becomes one 300-d vector from a frozen model: the mean of the top-layer states over non-pad positions, or the state at the last real token; causal models never let padding reach a real position (unit-tested). One multinomial logistic regression [12] on standardised features is fitted per representation, C chosen on the same 2,000 held-out training documents at every label-set size, the 7,600-document test split never used for selection (scaler and TF-IDF vocabulary are fitted on all 20,000 unlabelled training documents). Embedding-only baselines (the mean of the raw table rows) and a TF-IDF reference over the same vocabulary ids isolate what the sequence model adds.

Observations. *Pooling.* Mean pooling beats the last state for both the LSTM (67.9 vs. 62.3 macro-F1) and the Transformer (77.6 vs. 73.6).

Models. With mean pooling the Part I models rank 1-layer Transformer (78.9) > 4-layer Transformer (77.6) > RNN (73.5) > LSTM (67.9); 95% bootstrap intervals over the test documents are ±0.7–1.0 points and every gap quoted here is separable (1L−4L: +1.3 [+0.4, +2.1]), with the caveat of one LM seed per model. The best language model is the worst feature extractor.

Embedding variants. The LSTM with a frozen GloVe table gives 82.3 and with frozen skip-gram 79.3, far above the random table, but the raw tables alone do better still (GloVe 87.9, skip-gram 83.4), and even the LSTM's own learned input table (75.1) beats the LSTM's hidden states.

Data size (Fig. 4b). TF-IDF is the best representation with 20,000 labels (89.9) but falls to 78.3 with 500,

true \ pred	World	Sports	Business	Sci/Tech
World	**1570**	102	145	83
Sports	68	**1762**	30	40
Business	140	52	**1430**	278
Sci/Tech	102	51	248	**1499**

True	Pred.	Conf.	Text (truncated)
World	Sports	1.00	Maddux Wins No. 302, Baker Wins No. 1,000 Greg Maddux pitched the Chicago Cubs into the lead
World	Business	1.00	Oil Prices, Earnings Send Stocks Lower NEW YORK - Investors sent stocks lower Tuesday as oil
World	Sci/Tech	1.00	Ancient fungus 'revived' in lab Fungus from a deep-sea sediment core that is hundreds of
Business	World	1.00	World briefs LONDON - A man wielding a machete and a knife attacked two security guards at the
Business	Sci/Tech	1.00	Google puts desktop search privacy up front Google has announced a new desktop search

Table 5: Confusion matrix (left) and the most confident error of each distinct true/predicted pair (right) of the best LM-based representation (LSTM + frozen GloVe, mean).

where mean GloVe still gives 85.6: dense pre-trained features dominate the low-label regime, sparse lexical features win with plentiful labels, and every LM-based representation sits below the embedding-only baseline built from its own input table at every label-set size (at 20,000 labels the states lose 1.6–7.2 points relative to their own tables).

Errors (Table 5, for LSTM + frozen GloVe, mean). Business and Sci/Tech are confused most often (526 of 7,600 test documents), then World and Business (285); the most confident errors are partly label noise (sports and stock-market stories filed under *World*) and partly genuine ambiguity (a Google privacy story labelled Business, predicted Sci/Tech). Vocabulary coverage does not explain errors: the `<unk>` rate is 7.5% in correctly and 7.4% in wrongly classified documents.

Does the LM ranking agree with the downstream ranking? Only when the input table changes. Over all 6 LM-based mean-pooled representations $\rho(\text{PPL, F1}) = -0.60$ ($p = 0.21$; negative = agreement), but that agreement is carried entirely by the embedding table: across the 4 architectures sharing the random table $\rho = +0.40$ ($p = 0.60$; descriptive at this n), the best LM is the worst feature extractor and the 1-layer Transformer (a worse LM) the best, whereas across the 3 LSTM input-table cells $\rho = -1.00$: both metrics improve together, driven by the table, and the LSTM's states keep only part of what the table provides (82.3 vs. 87.9 for the raw GloVe mean). Our explanation, a hypothesis consistent with all three parts: a next-token objective rewards local syntax and the last few words, so sequence-model states are specialised away from topic; every model's mean-pooled states score *below* its own input table, by least for the Transformers (1.6 and 2.6 points, whose residual stream carries the embeddings to the top layer) and most for the LSTM (7.2), whose gates overwrite them. Two confounds remain: recurrent models were trained with state carried across the stream but Part III starts every 45-token document from a zero state, a cold start the Transformer never faces; and GloVe-6B's Gigaword newswire is a domain match for AG News that favours the raw GloVe features. Domain shift presumably hurts every LM-based representation, which we did not test; fine-tuning was outside the frozen-representation question.

6 Conclusions, limitations and use of AI tools

Across architectures, better next-token prediction did *not* give better frozen representations; across input tables both improved together, so the embedding table mattered more than the sequence model for the downstream task (for perplexity, architecture still dominated among usable tables). The LSTM is the best full-history predictor because it carries state beyond 128 tokens; at equal context the 4-layer Transformer is better, at 35% more training time and 80% of the LSTM's throughput on the common workload. A frozen table trained on the same 2 M tokens in 1.9 min beats jointly learned embeddings; GloVe's extra data adds 4.1 points, training the pre-trained table adds 7.3 more, the per-token gains are largest on rare words, and a rescaling control shows that the frozen GloVe table's gain over the trainable random table is input scale rather than content (among frozen tables, content still dominates). Downstream, mean-pooled raw GloVe (87.9) beats every LM-based feature and TF-IDF wins with enough labels (89.9). *Failure cases.* Greedy decoding degenerates into heading loops for every model (74% repeated 4-grams for the RNN); the frozen random table emits `<unk>` chains; the LSTM's last state is the worst representation (62.3); the best Part II cells loop more under greedy decoding, unexplained. *Lessons learned.* Evaluate under more than one context protocol before crediting an architecture; test the input table alone, and its scale, before crediting the sequence model; never read representation quality off perplexity. *Limitations.* One corpus, small models, a 12-epoch budget at which nothing converged; MPS is not bit-reproducible; GloVe-6B overlaps WikiText's sources and AG News's domain; Part III uses frozen models only; AG News has label noise. *Code and tools.* Code, configurations and results: `https://github.com/AniroodhChaudhary/dsa4213-assignment1`. The training loop follows PyTorch's `word_language_model` example; data via Hugging Face `datasets`. Claude Code (Anthropic, Claude Fable 5.1) [13] drafted the design, code, asset scripts and text edits; the author ran all experiments and owns the analysis (`AI_USAGE.md`).

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A All generated continuations

A.1 Part I models

Prompt 1 the americans knew of the approach of the japanese forces

KN 5-gram, greedy in the region . <eos> = = = = <eos> the first recorded in the inventory of <unk> <unk> , a <unk> , a <unk> , a <unk> , a <unk> , a <unk> , a <unk> , a <unk> , a <unk> , a <unk> ,

KN 5-gram, topk staged a sit in his home . <eos> with the rise of cobra commander in the sequel . <eos> he was appointed director @-@ <unk> during the late <unk> . another such cruise ship , loaded with troops , an individual 's perception have suggested that the team . the

RNN, greedy . <eos> = = = = = <eos> = = = = = <eos> = = = = = <eos> = = = = = <eos> = = = = = <eos> = = = = = <eos> = = = = = <eos> = = = = = <eos>

RNN, topk against the kpa 's king , and a peace @-@ force " it into a high position . <unk> had the right end of the 766th (" <eos> <unk> de <unk>) from the huai (emperor <unk>) in the cathedral area with the cathedral . <eos> the

LSTM, greedy . <eos> = = = = = = = = = = = = = = = <eos> the first time of the year , the first time , was the first time of the year , and the first time was the first time to be the first time

LSTM, topk during the conflict . <eos> the first time against the american union were created by the royal army , both the republic of their <unk> against the country , and one of the 1st century ce , although the general government was supported by his party . the first two

Transformer (4L), greedy . <eos> = <eos> the song was written by the song by the song , and the song was written

Transformer (4L), topk - and some of the song was " impossible " . <unk> said that " this were a " dramatic <unk> " and " <unk> ' and " <unk> " for the country ; the feast of their <unk> " was a standard word that was a major <unk> of

Transformer (1L, control), greedy . <eos> = = = = = <eos> the 766th regiment was the 766th regiment was the 766th regiment was the 766th regiment , and the 766th regiment was sent to reinforce the 766th regiment . the 766th regiment was sent to the 766th regiment , commanded by the 766th

Transformer (1L, control), topk against the 766th forces . <eos> the 766th headquarters was originally taken over them by the 766th regiment , 2nd division was appointed an <unk> unit of the <unk> , who had no advantage of fighting . these were lost after the 766th unit did not return to the 5th

Prompt 2 the ruins lie among the tropical rainforests of northern guatemala

KN 5-gram, greedy , and the <unk> of the <unk> , and the <unk> of the <unk> , and the <unk> of the <unk> , and the <unk> of the <unk> , and the <unk> of the <unk> , and the <unk> of the <unk> , and the <unk> of the <unk> ,

KN 5-gram, topk , and was replaced by the national library , which was a large @-@ scale <unk> paintings and drawings . a few months later a bomb . <unk> of <unk> , with the <unk> in an attempt to play in the fa cup , but the soviet archives of dumont

RNN, greedy , and <unk> , <unk> <unk> , <unk> <unk> , <unk> <unk> , <unk> <unk> , <unk> <unk> , <unk> <unk> , <unk> <unk> , <unk> <unk> , <unk> <unk> , <unk> <unk> , <unk> <unk> , <unk> <unk> , <unk> <unk> , <unk> <unk> , <unk> <unk> , <unk>

RNN, topk , and the middle section . <unk> for <unk> and the other group of the <unk> are used as a source of the georgian spacing is in the same . <eos> = = = early sources = = <eos> = = critical reception = = <eos> while critics were given

LSTM, greedy . the first two years later , the first of the first time was the first time in the first season , and the first time of the year , the <unk> of the <unk> , was the first time to be the first time . <eos> = = =

LSTM, topk , and the capital of the north end , which is not as a second <unk> and the east and west west of the southern gulf of the area , and it was the last five years of the united states during the first period , until september 14 ,

Transformer (4L), greedy . the north of the south of the east of the east of the south of the south korean peninsula , the south korean river , and the south korean peninsula were constructed in the north korean peninsula . the 766th was also built in the north korean area .

Transformer (4L), topk , the east 's western coast . the first few species of the west of <unk> were named as the site of the northern hemisphere and the north america , and south of <unk> and a small group of a. hygrometricus ; it was recorded . most recent species had

Transformer (1L, control), greedy , and the island of the island . <eos> = = = = = <eos> the storm was named after the storm , the storm was named after the storm , the storm was named after the storm , the storm was named after the storm , the storm was

Transformer (1L, control), topk , the island ’s cathedral . <eos> = = = = <eos> the east of <unk> (<unk>) = = = <eos> = = development = <eos> the new york times the first period = = <eos> the next year) = <eos> = = reception = = =

Prompt 3 in 1644 , hu took it upon himself to create

KN 5-gram, greedy a new york times , the song is a re @-@ interpretation of the hymn " joyful , joyful " , and " <unk> " . <eos> = = = = <eos> the first recorded in the inventory of <unk> <unk> , a <unk> , a <unk> , a <unk>

KN 5-gram, topk a map of the poem ’s the country ’s " little to no use of this mushroom as if not exactly jammed together . the new yorker , and later , already putting a bit more polished <unk> <unk> , was also the first time in his career . he

RNN, greedy a new <unk> , and the <unk> of the <unk> . <eos> = = = = = <eos> = = = = = <eos> = = = = = <eos> = = = = = <eos> = = = = = <eos> = = = = = <eos> = =

RNN, topk a suitable game . " , the player is the largest and other members of the united states as " its course " , and the belgian company , said in its existence of " one " <eos> <unk> of the city , " and the second of a year

LSTM, greedy a " <unk> " . <eos> = = = = = = = = = = = = = = = = <eos> = = = = = = = = = = = = = = = = = <eos> the first time of the film was the first time of

LSTM, topk a key of the school to the house of <unk> that was also used in the area . he also suggested that his life was to seek to a lot of education between the country and their <unk> <unk> , a number of six men . the <unk> @-@ off

Transformer (4L), greedy a <unk> of the jin . the song was the song by the jin , and the jin was the song . the song was the song of the song , and the song was the song by the song . the song was the song by the song song

Transformer (4L), topk a letter of the emperor and the jin would be attacked . the court of modern rule was forced to defend a jin . <eos> the huai river , although the jurchens attempted to secure a treaty , the song had been <unk> to hold the jin , this would

Transformer (1L, control), greedy a new state of the state of the state of the state of the state of the state of the state of the state of the state of the state of the state of the state of the state of the state of the state of the state of maryland

Transformer (1L, control), topk a budget of 5 @,@ 000 in 2011 . <eos> after the next year , he has been awarded a special candidate for the state of yugoslavia . <eos> there is no doubt that he may have the <unk> , as a <unk> of a year of <unk> in the

Prompt 4 as the highway approaches thompson , us 2 leaves the

KN 5-gram, greedy area . <eos> = = = = <eos> the first recorded in the inventory of <unk> <unk> , a <unk> , a <unk> , a <unk> , a <unk> , a <unk> , a <unk> , a <unk> , a <unk> , a <unk> , a <unk> , a <unk>

KN 5-gram, topk house and <unk> <unk> . the first four starts , and the <unk> <unk> of the hindu in a single . <eos> meanwhile , a 2 @-@ 3 . a total of 15 @,@ 000 @,@ 000 acres (2 @.@ 3 % in the area , and was an

RNN, greedy east of the <unk> <unk> . <eos> = = = <unk> = = = <eos> the kakapo is a <unk> of the <unk> <unk> (<unk>) , and <unk> (<unk>) , <unk> <unk> (<unk>) , <unk> <unk> (<unk>) , <unk> <unk> (<unk>

RNN, topk east state . <eos> = = = = tropical storm developed the season , a tropical wave the system peaked in the azores and a storm . the storm caused wind gusts to a high winds of 65 mph) . <eos> = = = = <eos> a large number

LSTM, greedy <unk> of the <unk> . <eos> = = = = <unk> = = = = <eos> the first part of the first season was the first time of the season , and the first time was the first time to be the first time . <eos> = = = =

LSTM, topk area in the first area of the final half . the first of the south atlantic ocean , an additional , <unk> bridge , is in the end of the northern atlantic ’s west . the road ’ north of the city is at the top the west and the

Transformer (4L), greedy east of the east coast of the united states . <eos> = = = route description = = <eos> the highway is a highway between the highway of the highway system . the highway is located in the north of the north of the north of the north of the

Transformer (4L), topk east) . <eos> = = = route description = = <eos> the route has been laid in a park , <unk> avenue and the state state route in the village of the east . the highway continues north and south . it is a few <unk> road in the

Transformer (1L, control), greedy highway . <eos> = = = = <eos> the highway is a highway system to the highway system . the highway was designated as the highway system as a highway system of the highway system . the highway was designated as the highway system . the highway was designated as

Transformer (1L, control), topk system to the first line and , with an early 20th century , and by a village , was completed . <eos> hakim is used in the united states and made from a second @-@ day @-@ old film . <eos> = = = = = <eos> = = =

Prompt 5 churchill , ward (1996) . from a native

KN 5-gram, greedy american tribes , the <unk> , the <unk> <unk> , the <unk> <unk> , the <unk> <unk> , the <unk> <unk> , the <unk> <unk> , the <unk> <unk> , the <unk> <unk> , the <unk> <unk> , the <unk> <unk> , the <unk> <unk> , the <unk> <unk> ,

KN 5-gram, topk plant , as they were on the next to the player . " <unk> " and that his personal health , bethesda , maryland , in its original form of the god apollo bunder in the uk still " . in a review for the <unk>) in a large

RNN, greedy american <unk> , <unk> , <unk> , <unk> , <unk> , <unk> , <unk> , <unk> , <unk> , <unk> , <unk> , <unk> , <unk> , <unk> , <unk> , <unk> , <unk> , <unk> , <unk> , <unk> , <unk> , <unk> , <unk> , <unk> , <unk>

RNN, topk america , which were not known as , <unk> , tintin , the <unk> of the dominican republic and ireland , frelimo , which was held on the site of the jin dynasty college of war ii (<unk>) . <eos> " this is <unk> by a new york

LSTM, greedy american academy , the <unk> of the <unk> , was the first time of the first time . the first time was the first time of the year , and the first time was the first time to be the first time . <eos> = = = = = =

LSTM, topk newspaper , he had not been built that the other troops in the state is " just so what they had happened of all the life and a] " . he concluded what it was born during the war . <unk> at a further <unk> from the british post

Transformer (4L), greedy american newspaper , the film was published in the united states . <eos> = = = = production = = <eos> the film was released in september 2011 . the film was released in september 2011 . the film was released in september 2011 . the film was released in

Transformer (4L), topk sentence , wheeler 's book was known as the " good and " the first year that year was not the finest of one of the best film of the united kingdom during 1996 . in 2000 of the united states , wheeler continued to <unk> his wife . he

Transformer (1L, control), greedy american academy (<unk>) , <unk> (<unk>) , <unk> (<unk>) , <unk> (<unk>) , <unk> (<unk>) , <unk> (<unk>) , <unk> (<unk>) , <unk> (<unk>) , <unk> (<unk>) , <unk> (<unk>)

Transformer (1L, control), topk korean , which were also described as " <unk> ") , a new school " (1999) . <eos> tintin in april 2010 , 2005 , " <unk> ") : written by james f. kennedy , and wrote that this period in <unk> that year . in

Model	Decoding	distinct-1	distinct-2	distinct-3	repeated 4-grams
KN 5-gram	greedy	0.24	0.36	0.39	0.58
KN 5-gram	topk	0.71	0.98	1.00	0.00
RNN	greedy	0.13	0.20	0.23	0.74
RNN	topk	0.64	0.91	0.97	0.01
LSTM	greedy	0.30	0.50	0.65	0.25
LSTM	topk	0.66	0.95	1.00	0.00
Transformer (4L)	greedy	0.30	0.48	0.61	0.31
Transformer (4L)	topk	0.63	0.94	0.99	0.00
Transformer (1L, control)	greedy	0.20	0.31	0.38	0.57
Transformer (1L, control)	topk	0.63	0.87	0.93	0.03

Table 6: Part I repetition statistics averaged over the five prompts.

A.2 Part II embedding settings (LSTM)

Prompt 1 the americans knew of the approach of the japanese forces

LSTM, random, trainable (Part I LSTM), greedy . <eos> = = = = = = = = = = = = = = = = <eos> the first time of the year , the first time , was the first time of the year , and the first time was the first time to be the first time

LSTM, random, trainable (Part I LSTM), topk during the conflict . <eos> the first time against the american union were created by the royal army , both the republic of their <unk> against the country , and one of the 1st century ce , although the general government was supported by his party . the first two

LSTM, random, frozen, greedy . the <unk> of the <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk>

<unk> <unk> <unk>

LSTM, random, frozen, topk have been observed and the <unk> . <unk> between the song 's actions had been developed by the <unk> with <unk> and the other <unk> - of the <unk> , who had little more for local . however , in december 2010 , ulysses said that the company of the

LSTM, skip-gram, trainable, greedy . <eos> = = = = <unk> = = = = <eos> the first year of the war , the first time of the war , was the first time of the war . <eos> = = = <unk> = = = <eos> the first day of the war ,

LSTM, skip-gram, trainable, topk such as he was a <unk> . the subsequent year of the 19th century . " one of the greatest <unk> " (as the eighth world war <unk> , also was based at the expense of men , in both the portuguese era) , and the <unk> .

LSTM, skip-gram, frozen, greedy . <eos> = = = = <unk> = = = = <eos> the first of the first two @-@ year @-@ old @-@ class @-@ class @-@ class <unk> was the first of the first class of the royal navy , and was the first of the first class of

LSTM, skip-gram, frozen, topk over the atlantic . <eos> the first three battalions of the 9th brigade division were sent from the south korean of south korean forces , taking it as <unk> . this was also supported to conduct the 766th . the 10th division sent forward into the battle of <unk> ,

LSTM, GloVe, trainable, greedy . <eos> = = = = <unk> = = = = <eos> the first of the first season of the first season was the first time in the first season . the first season of the season was the first time in the first season , and the first two

LSTM, GloVe, trainable, topk where it took it to the other . however , the next week they would have moved to the dominican and then were established , making for a <unk> . this was used by the soviets , although the british officer was elected against a number of people . <eos>

LSTM, GloVe, frozen, greedy . <eos> = = = = <unk> = = = = <eos> the first two @-@ year @-@ old <unk> , the <unk> , and the <unk> of the <unk> <unk> <unk> (<unk>) , and the <unk> <unk> (<unk> <unk>) , <unk> <unk> (<unk> <unk>

LSTM, GloVe, frozen, topk against the conflict . <eos> the first two chapters of the united kingdom made his own campaign in the 1920s , but he has a reputation of the <unk> . an attempt to ensure the assassination of independence , and also had taken down his own work and the "

Prompt 2 the ruins lie among the tropical rainforests of northern guatemala

LSTM, random, trainable (Part I LSTM), greedy . the first two years later , the first of the first time was the first time in the first season , and the first time of the year , the <unk> of the <unk> , was the first time to be the first time . <eos> = = =

LSTM, random, trainable (Part I LSTM), topk , and the capital of the north end , which is not as a second <unk> and the east and west west of the southern gulf of the area , and it was the last five years of the united states during the first period , until september 14 ,

LSTM, random, frozen, greedy . the <unk> of the <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk>

LSTM, random, frozen, topk , and a similar area of the same <unk> @-@ century , and the <unk> <unk> @-@ <unk> @-@ <unk> @-@ <unk> @-@ <unk> @-@ . <unk> <unk> , <unk> <unk> and <unk> <unk> by the <unk> <unk> <unk> , <unk> (<unk> <unk>) <unk> (" <unk> "

LSTM, skip-gram, trainable, greedy . <eos> = = = = <unk> = = = = <eos> the first time of the war was the first time of the war . <eos> = = = <unk> = = = <eos> the first day of the war was the first time of the war . the

LSTM, skip-gram, trainable, topk , and the modern sea of the southern <unk> in the united states . <eos> the kakapo 's range in south africa is part of a new season . <eos> in april 2009 , a female family was made on the newly published <unk> in <unk> (portuguese) and

LSTM, skip-gram, frozen, greedy . <eos> = = = = <unk> = = = = <eos> the first of the first two @-@ year @-@ old @-@ old @-@ old @-@ old @-@ <unk> <unk> , the <unk> of the <unk> , was the first of the first <unk> of the <unk> . <eos>

LSTM, skip-gram, frozen, topk , and the eastern mountains of the town , which is not not a well . by the late 20th century , the english navy ' <unk> , and <unk> , were the first best best for the most important . these had a few <unk> between six major groups

LSTM, GloVe, trainable, greedy . <eos> = = = = <unk> = = = = <eos> the first of the kakapo 's first season was the first time in the united states . the kakapo was the first @-@ largest ever ever ever ever ever ever , and the first time in the united

LSTM, GloVe, trainable, topk , and the western gulf of the island , which were at a 2 - 1 km (8 @.@ 7 mi) long @-@ distance , and in the north end , the majority of the population was also the oldest for the largest <unk> which came with a

LSTM, GloVe, frozen, greedy . <eos> = = = = <unk> = = = = <eos> the first @-@ term <unk> of the

<unk> <unk> <unk> <unk> (<unk> <unk>) , the <unk> <unk> (<unk> <unk>) , was a <unk> of the <unk> <unk> <unk> <unk> <unk> (<unk> <unk>)

LSTM, GloVe, frozen, topk , and the middle coast of the east , which is not an area of <unk> in the eastern eastern pacific of western africa . they were not known in the eastern south of north africa . the song 's <unk> fleet on the eastern north china area and its

Prompt 3 in 1644 , hu took it upon himself to create

LSTM, random, trainable (Part I LSTM), greedy a " <unk> " . <eos> = = = = = = = = = = = = = = = <eos> = = = = = = = = = = = = = = = = <eos> the first time of the film was the first time of

LSTM, random, trainable (Part I LSTM), topk a key of the school to the house of <unk> that was also used in the area . he also suggested that his life was to seek to a lot of education between the country and their <unk> <unk> , a number of six men . the <unk> @-@ off

LSTM, random, frozen, greedy the <unk> . <eos> = = = = <unk> = = = = <eos> the first @-@ year @-@ century was the first of the first time in the united states , and the first time of the united states was the first time of the united states . the

LSTM, random, frozen, topk a variety of the people and the jin that the 130th of the state was only a small period of their parents of the song . despite the second phase , they lived a short at their <unk> <unk> , a new <unk> and the <unk> . <eos> = =

LSTM, skip-gram, trainable, greedy a <unk> . <eos> = = = <unk> = = = <eos> the first time of the war was the first time of the war . <eos> = = = <unk> = = = <eos> the first day of the war , the 766th was the first time of the

LSTM, skip-gram, trainable, topk a car . <eos> = = = post @-@ war tour = = = <eos> " he had seen a special experience that was to be interested in the audience . this makes it been an ideal <unk> , and it is <unk> " . <eos> = = music =

LSTM, skip-gram, frozen, greedy the <unk> of the <unk> . <eos> = = = = <unk> = = = = <eos> the first two @-@ year @-@ old @-@ old @-@ old <unk> <unk> was the first of the first <unk> of the <unk> , and the first two @-@ year @-@ old @-@

LSTM, skip-gram, frozen, topk a brief of the government and the former president of their united states . <eos> = = = critical response = = = <eos> the first horizons of the treaty in 1992 began a plan for " <unk> <unk> " and " <unk> " and " to <unk> " .

LSTM, GloVe, trainable, greedy a <unk> of the <unk> of the <unk> . <eos> = = = <unk> = = = <eos> the first two years of the war was the first of the first two years of the war . the first of the first two years of the war was the first

LSTM, GloVe, trainable, topk a criminal in the kingdom of <unk> in may . when the english army was formed in an interview with finkelstein 's post , the first editor of the palestinian state has begun to stay on an <unk> <unk> , as a <unk> and a " <unk> <unk> " .

LSTM, GloVe, frozen, greedy the <unk> of the <unk> . <eos> = = = = <unk> = = = = <eos> = = = = <unk> = = = = <eos> = = = = <unk> = = = = <eos> = = = = <unk> = = = = <eos> = = =

LSTM, GloVe, frozen, topk a peace of the old family , who also were granted to the united states in august 1991 . the majority of the three people was arrested in the dominican and australia between their territory and became <unk> <unk> . the city , with a number of <unk> to the

Prompt 4 as the highway approaches thompson , us 2 leaves the

LSTM, random, trainable (Part I LSTM), greedy <unk> of the <unk> . <eos> = = = = <unk> = = = = <eos> the first part of the first season was the first time of the season , and the first time was the first time to be the first time . <eos> = = = =

LSTM, random, trainable (Part I LSTM), topk area in the first area of the final half . the first of the south atlantic ocean , an additional , <unk> bridge , is in the end of the northern atlantic 's west . the road ' north of the city is at the top the west and the

LSTM, random, frozen, greedy <unk> of the <unk> . <eos> = = = = <unk> = = = = <eos> the first @-@ language war of the song was the first time of the first season , and the first time in the united states . the first year was the first time in

LSTM, random, frozen, topk world as a <unk> . the game 's largest , was the first and most successful television , with a " <unk> race " and the <unk> . <eos> = = = = in the same : = = = = <eos> = = = = <unk> = = =

LSTM, skip-gram, trainable, greedy <unk> <unk> <unk> <unk> (<unk>) . <eos> = = = <unk> = = = <eos> the first time of the united states , the united states , was the first time of the united states . the first time was the first time of the war . <eos>

LSTM, skip-gram, trainable, topk line to the <unk> . the two other groups . the <unk> <unk> <unk> for the past <unk> of the <unk> <unk> al @-@ <unk> <unk> <unk> the <unk> the the <unk> . <eos> in a week on 7 august , the congo 's <unk> of <unk> , the <unk>

LSTM, skip-gram, frozen, greedy <unk> <unk> , and the <unk> <unk> <unk> <unk> <unk> <unk> <unk>

<unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk>

LSTM, skip-gram, frozen, topk state 's <unk> <unk> . the two @-@ foot <unk> to the <unk> <unk> and the o <unk> was made in <unk> province and the first of the first year as a single of the second century in the early 19th century . the first year the war was to

LSTM, GloVe, trainable, greedy <unk> , <unk> , <unk> , <unk> , <unk> , <unk> , <unk> , <unk> , <unk> , <unk> , <unk> , <unk> , <unk> , <unk> , <unk> , <unk> , <unk> , <unk> , <unk> , <unk> , <unk> , <unk> , <unk> , <unk> , <unk> ,

LSTM, GloVe, trainable, topk east to the <unk> . the most common approach of the " <unk> " (" [<unk>]) and <unk> means " (<unk> . the <unk>) are an attempt to be used as " <unk> " , and is not a common <unk> . <unk> the

LSTM, GloVe, frozen, greedy <unk> of the <unk> <unk> . <eos> = = = = <unk> = = = = <eos> the first @-@ class @-@ class @-@ class @-@ class @-@ class @-@ class @-@ class @-@ class @-@ class @-@ class @-@ class @-@ class @-@ class @-@ class @-@ class @-@

LSTM, GloVe, frozen, topk national navy . <eos> = = = = final of the first two two weeks = = = = <eos> the <unk> sector 's history of the world and the uk is a number of people in that they were to be seen in the west <unk> . the first

Prompt 5 churchill , ward (1996) . from a native

LSTM, random, trainable (Part I LSTM), greedy american academy , the <unk> of the <unk> , was the first time of the first time . the first time was the first time of the year , and the first time was the first time to be the first time . <eos> = = = = = =

LSTM, random, trainable (Part I LSTM), topk newspaper , he had not been built that the other troops in the state is " just so what they had happened of all the life and a] " . he concluded what it was born during the war . <unk> at a further <unk> from the british post

LSTM, random, frozen, greedy of the <unk> , the <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk>

LSTM, random, frozen, topk level , it is a large @-@ level <unk> , which was used to be an unusual that they have be collected . it was also described as a " <unk> " system system in the 1950s ; the song is an extensive at the <unk> church in the same

LSTM, skip-gram, trainable, greedy american <unk> , the <unk> of the <unk> , <unk> , <unk> , <unk> , <unk> , <unk> , <unk> , <unk> , <unk> , <unk> , <unk> , <unk> , <unk> , <unk> , <unk> , <unk> , <unk> , <unk> , <unk> , <unk> , <unk> , <unk>

LSTM, skip-gram, trainable, topk edition , he has his father and her own name for the end of the world war i and his life until the month , at a day . the family will fly towards the school : she was a strong school at his age <unk> by their wife .

LSTM, skip-gram, frozen, greedy american war , the <unk> of the <unk> <unk> <unk> , was a <unk> of the <unk> <unk> <unk> , <unk> <unk> , <unk> <unk> , <unk> <unk> , <unk> <unk> , <unk> <unk> , <unk> <unk> , <unk> <unk> , <unk> <unk> , <unk> <unk> , <unk> <unk> ,

LSTM, skip-gram, frozen, topk treaty , he became in the united states , and made the second two years in december 1914 . the first anniversary of the first war was a long @-@ long plan against madero in the 1920s with the new royal military police by the <unk> river of the british

LSTM, GloVe, trainable, greedy american <unk> , the <unk> <unk> <unk> <unk> <unk> <unk> (<unk> <unk>) , <unk> <unk> (<unk> <unk>) , <unk> <unk> (<unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk> <unk>

LSTM, GloVe, trainable, topk review , david finkelstein " 's the most <unk> of her , " , for his arrival on a book , criticized the organization of " a book in the american christian literature ... " of many scholars . <eos> = = background = = <eos> following his death of

LSTM, GloVe, frozen, greedy american <unk> , the <unk> of the <unk> <unk> (<unk>) , <unk> <unk> (<unk>) , <unk> (<unk> <unk>) , <unk> (<unk> <unk>) , <unk> (<unk> <unk>) , <unk> <unk> (<unk> <unk>) , <unk> <unk> (<unk> <unk>)

LSTM, GloVe, frozen, topk portion , he had not been described as the " common <unk> " . the second generation was used as the sole of a " <unk> " of the new york class line - " the greatest extent of the war " and his own <unk> from the united states

Embedding setting	Decoding	distinct-1	distinct-2	distinct-3	repeated 4-grams
random, trainable (Part I LSTM)	greedy	0.30	0.50	0.65	0.25
random, trainable (Part I LSTM)	topk	0.66	0.95	1.00	0.00
random, frozen	greedy	0.20	0.32	0.40	0.56
random, frozen	topk	0.54	0.84	0.93	0.04
skip-gram, trainable	greedy	0.23	0.39	0.50	0.41
skip-gram, trainable	topk	0.65	0.90	0.97	0.00
skip-gram, frozen	greedy	0.22	0.40	0.52	0.41
skip-gram, frozen	topk	0.63	0.91	0.98	0.00
GloVe, trainable	greedy	0.22	0.36	0.47	0.46
GloVe, trainable	topk	0.66	0.95	0.99	0.00
GloVe, frozen	greedy	0.20	0.36	0.46	0.46
GloVe, frozen	topk	0.66	0.92	0.98	0.00

Table 7: Part II repetition statistics averaged over the five prompts.

B Embedding nearest neighbours

Word	Skip-gram (own, WikiText-2 train)	GloVe-6B
`king`	`veera, leinster, ballala, throne, vassal`	`queen, prince, monarch, kingdom, throne`
`city`	`metropolitan, borough, town, conurbation, burnside`	`cities, town, downtown, mayor, area`
`war`	`outbreak, airbase, italo, civil, hawaiians`	`wars, conflict, battle, civil, occupation`
`album`	`daydream, yankovic, charmbracelet, rca, slayer`	`albums, song, ep, band, songs`
`1990`	`1989, pr, oceania, sap, pursuant`	`1991, 1992, 1987, 1989, 1986`
`north`	`south, west, east, southward, alaska`	`south, east, korea, northeast, west`
`played`	`playing, striker, staged, matches, shortstop`	`playing, plays, play, player, matches`
`@-@`	`93rd, 10m, amassing, defamation, dash`	`channels, regions, beneficial, distress, cable`

Table 8: Five nearest neighbours (cosine) of probe words in the two pre-trained tables.

C Full hyper-parameters and reproduction

Setting	Value
Embedding dim / Transformer d_{model}	300
RNN / LSTM	2 layers, hidden 300, tanh / LSTM cells
Transformer	4 layers (control: 1), 6 heads, FFN 1200, pre-LN, GELU, learned positions, context 128
Dropout	0.3 after embedding, between layers, before output
Batch × BPTT	32 × 128 tokens (continuous stream, state detached between chunks)
Optimiser	Adam, lr 0.001, β=(0.9, 0.98), 200-step linear warm-up
Schedule	halve lr after 2 consecutive epochs without val. improvement; early stop after 3; max 12 epochs
Gradient clipping	global norm 1.0
Output layer	untied (frozen input tables would otherwise freeze the softmax)
Validation / test protocol	recurrent: full history, batch 1; Transformer: sliding window 128 / stride 32
Decoding	greedy and top-k=40, T=1.0, 50 new tokens, seed 1234+prompt index
n-gram	interpolated Kneser–Ney, orders 2–5 fitted, order chosen on validation, $D_n = n_1/(n_1 + 2n_2)$
Skip-gram	gensim, sg=1, window 5, 10 negatives, subsampling 10^{-4}, 10 epochs, min_count 1, 1 worker
GloVe	6B tokens, 300d, uncased; missing rows $\mathcal{N}(0, \sigma_d)$ from the found rows
Part III TF-IDF reference	scikit-learn TfidfVectorizer over vocabulary ids: uni+bigrams, min_df 2, sublinear tf, fitted on the 20,000 training documents
Part III classifier	scikit-learn multinomial logistic regression (lbfgs), standardised features, $C \in \{0.01, 0.1, 1, 10\}$ on validation
Seeds	0 everywhere; Part II required settings also seed 1

Table 9: Every setting shared by the runs; per-run differences are the model/embedding blocks in `configs/`.

Reproduction: the repository `https://github.com/AniroodhChaudhary/dsa4213-assignment1` contains all code, configurations and results; `run_all.sh` re-creates every number (about 7 h on the M4 with the 24-epoch runs, 5 h without); `README.md` documents data access, preprocessing, dependencies, commands and seeds.